

- explain the concept of data generation process; and
- transform intrinsically linear models and interpret them.

1.1 INTRODUCTION

Econometrics is a specialised branch of economics which is concerned mostly with empirical verification of economic theory. Let us see how econometrics is different from other branches of economics such as mathematical economics or economic statistics. Mathematical economics usually presents economic theory in mathematical form. It does not bother about empirical measurement of economic theory. Econometrics on the other hand is mainly concerned with empirical verification of economic theory. In doing so, econometrics makes use of the mathematical equations suggested in mathematical economics. Economic statistics deals with collection, processing and presentation of data. Statistics provides many theories on the basis of which inferences can be drawn. For example, the probability rules and sampling distribution are quite important in hypothesis testing. The application of these rules to economic theory through empirical research is a part of econometrics.

In recent years theoretical developments in econometrics have been quite rapid. In fact, it is somewhat difficult to keep track of all these developments. The present unit serves as a foundation for the course. It begins with certain basic concepts and describes how econometric methods are applied to real life situations.

1.2 NATURE OF ECONOMETRICS

Econometrics is a branch of economics which deals with economic measurement. It may be defined as ‘the quantitative analysis of actual economic phenomena based on the concurrent development of theory and observation, related by appropriate methods of inference’ [Samuelson, Koopmans and Stone, 1954]. We should notice three issues being emphasized in the above statement. First, econometrics deals with quantitative analysis of economic relationships. Second, it is based on economic theory and logic. Third, it requires appropriate methods to draw inferences.

1.2.1 Theoretical Econometrics and Applied Econometrics

Econometrics has both theoretical side as well as applied side. Development of theoretical econometrics has been on two lines: (i) estimation methods, and (ii) test statistics. There are three broad estimation methods: (i) least squares method, (ii) maximum likelihood method, and (iii) method of moments. Under each of these methods, we have several estimation techniques. The least squares method was officially discovered and published by Legendre in 1805 and Carl Friedrich Gauss (1809) made substantial contribution to theoretical advancement of the least squares method. Under least squares method we have many estimation methods such as ordinary least squares, weighted least squares, generalised least

squares, indirect least squares, non-linear least squares, etc. We will discuss some of these methods in later Units of this course.

In applied econometrics, we apply these theoretical methods and test statistics to data obtained from various branches of econometrics (such as industrial organisation, labour economics, financial economics, etc.). Very often, the distinction between theoretical econometrics and applied econometrics is blurred – there are several studies which have developed new econometric tools and applied it to empirical data.

Through logic and experience economists have attempted to establish relationship between several variables. Movements in certain economic variables are explained in terms of changes in some other variables. For example, the law of demand states that as price increases the quantity demanded of a commodity decreases. As price of oilseeds increase farmers allocate larger land area for cultivation of oilseeds. You will find numerous examples of such relationships. Some of these are apparent while others are quite complex. As we do not know what exactly is happening we often make conjectures based on logic and our understanding of things.

Due to differences in perception, often different researchers explain the same phenomenon differently. For example, historical data on macroeconomic variables of a country over a period of time are the same. Inferences drawn by researchers on the basis of the same data could be different! While a researcher with classical ideas may come up with the view that government intervention is ineffective, another researcher with Keynesian ideas may suggest the need for government intervention. Thus, the variables included in a model, the nature of relationship among variables, the assumptions made, etc. are quite important.

Some of the economic relationships have been proved through empirical studies and validated under several situations. Some more are put forth as hypothesis and unequivocal conclusions are yet to be drawn. In both the cases, however, logic needs to be supported by empirical observation. An advantage with the subject matter of economics, unlike many other disciplines in social sciences, is that many of the economic variables can be quantified. This has helped in empirical measurement of economic variables and validation of economic theory.

In applied econometrics we deal with empirical studies based on a sample. As you know, it is not possible to carry out a census due to constraints on budget, time and personnel requirements. We consider a single sample and estimate the econometric model (for example, a regression equation). Thus, the population regression function and sample regression function are different. From the estimates obtained from a sample we try to draw inferences about the population. Thus, statistical significance of the estimates is very important in econometric models.

1.2.2 Econometric Models

In economics we often use the term ‘model’. It refers to a simplified version of reality. It allows us to understand, analyse and predict economic behavior. An

economic model can be for a microeconomic agent such as household or firm. For analysis of production behavior of a firm we assume the existence of production function such as Cobb-Douglas or translog, which determines our supply curve. We assume the structure of the market (perfect competition or oligopoly, for example), which decides the demand curve. By combining the supply curve and the demand curve, we obtain estimates of equilibrium price, output, capacity utilization, etc.

In macroeconomics, economic models represent the behavior of the economy as a whole. Here, we identify relevant macroeconomic variables (such as income, output, expenditure, investment, saving, exports, etc.) and establish a relationship among them. The relationships among these variables may be expressed through diagrams or mathematical equations. There could be macroeconomic models without mathematical expressions, but these may not be precise.

An economic model is based on certain assumptions. These assumptions are required so that minute details are ignored and essential elements are included. Let me illustrate the point through an example. In the case of a firm, it is assumed that there are two factors of production, viz., capital and labour. We combine all types of labour into a homogeneous category – we do not distinguish between a manager and a worker in the field! Similarly, while describing an indifference curve we overlook the type of households – the behavior of a rich household would be different from that of a poor household; or the behavior of a household in a rural area would be different from that of a household in an urban area. We ignore such details because our objective is to analyse the behavior of households to changes in prices and income. If our objective is to identify the changes in consumption pattern across households, we would require a different model and consider such differences.

In the Keynesian model, to take an example from macroeconomics, we consider aggregate variables such as total consumption, total investment, government expenditure, and net exports. We determine equilibrium level of output for the economy as whole. We ignore the behavior of households and firms. Several growth models (such as Harrod-Domar model or Solow model) assume that the economy consists of just one sector – there is an aggregate production function, which gives the relationship between aggregate output (that is, total output) with aggregate inputs (that is, total capital and total labour). It may sound unrealistic, but the objective of these growth models is to analyse the equilibrium conditions for economic growth, saving ratio and population growth. These models ignore the details, but the broad conclusions drawn are helpful in policy formulation. A question such as, ‘why growth rate differs across countries?’ can be addressed through these growth models.

1.2.3 Important Steps in an Econometric Study

Every econometric study is different in the sense that its objective, methodology, and database are different. The variables included in a study and the methods of estimation of parameters vary across studies. There are, however, certain

procedures and sequence which all econometric studies follow. Let us discuss the important steps involved in empirical studies applying econometric methods.

- a) The first and foremost issue in econometrics is the statement of hypothesis. Hypothesis is an assertion or claim made about a property or characteristic of a population. Thus, hypothesis is a tentative statement about the research questions of the study. It could be in the form of a simple statement or it could be in the form of relationship among variables. Hypothesis is drawn from economic theory or some logic built by the researcher.
- b) The second step is transformation of the above hypothesis into mathematical equation(s). This is our economic model that we want to study. Hypothesis is stated in the form of 'null hypothesis' and 'alternative hypothesis'. Null hypothesis, as you know, present a statement to the effect that there is no relationship between variables.
- c) The third step is the collection of relevant data required for the study. Data could be from primary or secondary sources. Usually, data are entered into a worksheet (such as an excel sheet). Data pertains to the variables included in the model.
- d) Estimation of parameters of the economic model. This requires selection of appropriate method of estimation. Broadly, there are three methods of estimation: (i) least squares method, (ii) maximum likelihood method, and (iii) method of moments. We will discuss about these methods in subsequent Units. Based on the probability structure of the data available to us, we select the appropriate estimation method.
- e) Testing of the hypothesis mentioned in the first step above. Such testing is based on the value of test statistic (such as t-ratio) or the p-ratio. Basically, it amounts to statistical significance of the estimates obtained by us.
- f) Interpretation of the economic model. On the basis of inferences drawn through hypothesis testing, we interpret our results.

Check Your Progress 1

- 1) What is an econometric model?

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- 2) Distinguish between theoretical econometrics and applied econometrics.

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- 3) Write down the important steps to be followed in an econometric study.

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1.3 SPECIFICATION OF A REGRESSION MODEL

In economic theory we attempt to establish relationship among variables. Such relationship could be of two types: deterministic and stochastic. Let us focus on the case of two variables: Y and X . As you know from your statistics course, the variables in a regression model are asymmetrical – one variable is dependent (Y) while the other is independent (X).

1.3.1 Deterministic Relationship

The simplest relationship between two variables, Y and X , could perhaps be a linear *deterministic* function given by

$$Y_i = \beta_0 + \beta_1 X_i \quad (i = 1, 2, \dots, n) \quad \dots (1.1)$$

In the above equation, X is the explanatory variable (or, independent variable) while Y is the explained variable (or, dependent variable). You should note that the subscript i represents the observation number, i ranges from 1 to n , the sample size. Thus, Y_1 is the first observation of the dependent variable, X_1 is the first observation of the independent variable, X_5 is the fifth observation of the independent variable, and so on.

Equation (1.1) implies that Y is completely determined by X and the parameters α and β . Suppose we have parameter values $\beta_0 = 3$ and $\beta_1 = 0.75$, then our linear equation is $Y_i = 3 + 0.75 X_i$. From this equation we can find out the value of Y for given values of X . For example, when $X = 8$, we find that $Y = 9$. Thus, if we have different values of X , then we obtain corresponding Y values on the basis of (1.1). Further, if X_i is the same for two observations, then the value of Y_i will also be identical for both the observations. A plot of Y on X will show no deviation from the straight line with intercept β_0 and slope β_1 .

If we look into the deterministic model given by (1.1) we find that it may not be appropriate for describing economic interrelationship between variables. For example, let Y = consumption and X = income of households. Suppose you record your income and consumption for successive months. For the months when your income is the same, does your consumption remain the same? Certainly not, because there are many other factors that decide your monthly consumption expenditure. The point we are trying to make is that economic relationships involve certain randomness. We cannot include all the determinants of monthly consumption expenditure in a regression model. For the same X_i , we observe different Y_i values.

1.3.2 Stochastic Relationship

The word ‘stochastic’ means ‘random’ or ‘involving certain probability’. Accordingly, stochastic relationship between X and Y involves certain randomness. In equation form it is given by

$$Y_i = \beta_0 + \beta_1 X_i + u_i \quad \dots (1.2)$$

where u_i is the stochastic error term. In equation (1.2) for the same value of X we obtain different values of Y due to the presence of the variable u . In the income consumption relationship, two individuals having the same income can have different expenditure. Stochastic relationship describes real life situation more accurately. Two employees having the same monthly salary in an office can have different monthly expenditure levels. One employee may have a bigger family to support, may have an ailing mother, or an altogether different lifestyle!

Remember that the right-hand side of (1.2) has two parts, viz., i) deterministic part (that is, $\beta_0 + \beta_1 X_i$), and ii) stochastic or randomness part (that is, u_i). Equation (1.2) implies that even if X_i remains the same for two observations, Y_i need not be the same because of different value of u_i . An implication of (1.2) is that if we plot (X, Y) on a graph paper the observations will not remain on a straight line.

It is assumed that the stochastic error term follows certain assumptions:

$$(i) \quad E(u_i) = 0 \quad \dots (1.3)$$

While the error is positive for some observations, it is negative for the others. It implies that on an average, the error is zero. In other words, the sum of errors across the population is zero.

$$(ii) \quad E(u_i^2) = \sigma^2 \quad \dots (1.4)$$

The variance of the error variable is σ^2 . It is constant across the population. For example, the variance of household consumption expenditure is the same for rich households as well as poor households.

$$(iii) \quad E(u_i u_j) = 0 \text{ for } i \neq j \quad \dots (1.5)$$

There is no correlation between the error terms for different observations in a dataset.

$$(iv) \quad E(X_i u_i) = 0 \quad \dots (1.6)$$

The explanatory variable is not correlated with the error term. This assumption is required for fulfilment of the properties of the estimators ($\hat{\beta}_0, \hat{\beta}_1$) on unbiasedness and consistency. You will learn more about the relevance of this assumption in Unit 4.

$$(v) \quad u_i \sim N(0, \sigma^2) \quad \dots (1.7)$$

The error variable u_i follows normal distribution with zero mean and σ standard deviation. For a given value of X , there is a whole range of values of u and a particular value is selected randomly.

The above assumptions are called the classical assumptions. These assumptions indicate the ideal conditions regarding the error variable. In empirical studies, some of the above assumption may not be fulfilled. We will discuss the implications of these assumptions in Unit 4.

1.3.3 Inclusion of Error Variable

In the previous sub-section we discussed about the statistical properties of the error variable (u_i). Now, let us find out the logic behind inclusion of the error variable in the regression model. There are at least three reasons for inclusion of the stochastic error term in a regression model.

First, in real life the relationship among variables is complex. Regression model is a simplification of such relationships. It is not always possible to include all relevant variables in the functional relation. For example, we assume that price is the sole determinant of demand for a product. In fact, several variables related to demand, such as individual tastes, population, income, etc. do influence the price of a product, which are omitted from the function. Some of these variables have positive effects while others have negative effects. The net effect of these omitted variables in the equation is represented by the error term. If the effect of these omitted variables is small, it is reasonable to assume that the error term is random.

Second, we do not know the exact functional form. Suppose we hypothesize a linear relationship among the variables, while it is a quadratic one. There is no means of knowing what the true relationship is. We just take one functional form on basis of our logic and reasoning. Certain error creeps into the model, because of the wrong specification of the functional form. Presence of an error term absorbs all such misspecification issues.

Third, another source of error is the sampling error. For instance, consider equation (1.2) as a household-consumption function, where Y is consumption and X is income. Even if equation (1.2) is a correct relationship, the sample we randomly choose to examine may turn out to be predominantly poor families. Thus, estimates of β_0 and β_1 from this sample group may not be as good as the estimates obtained from a representative sample.

1.4 DATA GENERATION PROCESS

By now you must be clear that econometric studies require specification of the model in quantitative terms, involving variables and parameters. In this context, selection of relevant variables and specification of the functional form are quite important.

Let us examine the consumption-income model to illustrate the above point. We observed that monthly consumption expenditure of two households having the same monthly income (say, Rs. 50000) are different – while one household consumes Rs. 35000, the other household consumes Rs. 48000. After considering its income level and other determinants of household expenditure, a household decides on its consumption level. What are these other determinants of household

expenditure? These could be family size, health status, culture, tastes, etc. all these determinants collectively determine the consumption expenditure of a household. Do we know all the relevant variables? Is it possible to include all these variables in our regression equation?

While specifying a consumption function, it is not possible to include all the determinations of consumption expenditure. Further, as pointed out in Sub-Section 1.2.2, an econometric model is a simplification of reality. Thus, in an econometric model we include the important variables only.

From our sample survey (or, from the secondary data sources) we obtain data. A major issue before us is our ignorance of the data generation process (DGP). Such data can be seen as a joint probability distribution with certain probability distribution. As mentioned above, we do not know all the factors that influence the consumption decision of a household. We make some guesswork, based on our experience and logic. Accordingly, we assume a probability structure for the data.

Data generation process (DGP) reflects the reality that is taking place. As DGP is not known to us, we try to uncover the secrets through our econometric models. There are two approaches to econometric modelling: (i) general-to-specific, and (ii) specific-to-general. In the first approach we consider a general model by including many variables. Subsequently, we exclude the irrelevant variables. In the second approach, we begin with a simple model and keep on adding variables based on our logic. Although, there is no consensus on the approach to be followed, researchers prefer the general to specific approach.

There are four steps in the process of building an econometric model: (i) we guess the data generation process, (ii) we assume the associated probability theory, (iii) we use that theory for empirical evidence, and (iv) we revise the model if the results do not match our data. We continue with the above four steps in an iterative manner till we get satisfactory results.

We make several trial and error on the functional form. If the data fails to fit our model, it implies we do not understand the data generation process. We try with another model. If the data fits our model, we still do not understand the data generation process! It just implies that our model was good in this case.

You should recall how a hypothesis is formulated. It is a statement based on theory or logic. If we could reject a null hypothesis, we confirm that the dataset does not hold true (recall Karl Popper's theory of falsification discussed in MEC 109).

1.5 FUNCTIONAL FORMS

Usually, we consider a linear regression equation in our econometric model, as it is easier to estimate and interpret such models. Linear regression models are linear in parameters and linear in variables. Linear regression models, however, may not be appropriate in all cases; there could be situations where the regression model is non-linear.

1.5.1 Classical Regression Model

The two-variable linear regression model is the simplest functional form that we consider. It is given by

$$Y_i = \beta_0 + \beta_1 X_i + u_i \quad \dots(1.2)$$

where Y is the explained variable (also called dependent variable)

X is the explanatory variable (also called independent variable)

u is the error variable (also called white noise in time series analysis)

When we include more than one explanatory variable in the regression model, we call it the multiple regression model. In notations, a multiple regression model is given by

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k + u \quad \dots (1.8)$$

In equation (1.8) there is one explained variable (Y) and k explanatory variables ($X_1, X_2, \dots, X_k$). We have omitted the sub-scripts for simplicity.

When the error variable u fulfils all the classical assumptions specified at equations (1.3) to (1.7), we call the regression model a 'classical regression model'. Ordinary Least Squares (OLS) method is frequently used to estimate parameters of the classical regression models.

1.5.2 Intrinsically Linear Cases

Certain non-linear equations can be transformed into linear equations by taking logarithms. Two intrinsically linear models are used frequently: (i) semi-log model, and (ii) log-linear models.

(a) Semi-Log Models

Let us begin with a functional form as follows:

$$Y_t = e^{\beta_0 + \beta_1 X_t + u_t} \quad \dots (1.9)$$

This regression model, in its present form, is non-linear. Therefore, it cannot be estimated by OLS method. However, if we take natural logs of both the sides, we obtain

$$\ln Y_t = \beta_0 + \beta_1 t + u_t \quad \dots (1.10)$$

It transforms into a semi-log equation. It is called a semi-log model as one of the variables is in log form.

If we take $\ln Y_t = Y'_t$, then equation (1.10) can be written as

$$Y'_t = \beta_1 + \beta_2 t + u_t \quad \dots (1.11)$$

Estimation of equation (1.11) is simple. The equation is linear in parameters and in variables. Thus, we can apply OLS method to estimate the parameters. The implication of the regression model (1.11), however, is much different from the regression model (1.2).

If we take the differentiation of the regression model $Y_t = \beta_0 + \beta_1 X_t + u_t$, we obtain

$$\frac{dY}{dt} = \beta_1 \quad \dots (1.12)$$

Equation (1.12) shows that the slope of the regression equation is constant. An implication of the above is that the absolute change in the dependent variable for unit increase in the independent variable is constant throughout the sample. If there is an increase in X by one unit, Y increases by β_1 unit.

Now, let us consider the regression model $\ln Y_t = \beta_0 + \beta_1 t + u_t$. If we take differentiation of equation (1.11) we find that

$$\frac{d \ln Y_t}{dt} = \beta_1$$

which means

$$\frac{1}{Y_t} \frac{dY_t}{dt} = \beta_1 \quad \dots (1.13)$$

An implication of equation (1.13) is that the slope of the regression model is variable. Thus, its interpretation is different from that of the regression model $Y_t = \beta_0 + \beta_1 X_t + u_t$.

For equation (1.13), we interpret the slope coefficient (β_1) as follows: For every unit increase in X , there is β_1 *per cent* increase Y . Thus, for a semi-log model the change in the dependent variable in terms of percentages. The semi-log model is useful in estimating growth rates.

(b) Log-Linear Models

Let us consider the following non-linear regression model

$$Y = \beta_0 X^{\beta_1} \quad \dots (1.14)$$

This model will be intrinsically linear if it can be transformed into

$$Y' = \beta_0 + \beta_1 X' + u \quad \dots (1.15)$$

Using the logarithm of each of the variable in equation (1.13), we get the following transformed equation:

$$\ln Y_i = \beta_0 + \beta_1 \ln X_i + u_i \quad \dots (1.16)$$

The regression model given at (1.16) is called a log-linear model (because it is linear in logs of the variables) or double-log model (because both variables are in log form).

Let us take differentiation of equation (1.16) with respect to $\ln X_i$

$$\frac{d(\ln Y_i)}{d(\ln X_i)} = \beta_1 \quad \dots (1.17)$$

A closer look at equation (1.17) shows that the slope parameter of (1.14) represents the elasticity between Y and X . This attractive feature of the log-linear model has made it popular in applied work. The slope coefficient β_1 measures the elasticity of Y with respect to X , that is, the percentage change in Y for one per